Issac Roy

U.S. Citizen, San Diego, CA

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Education

University of California San Diego (3.97 GPA) **March 2027**

B.S. Data Science | *Math Minor* | *Cognitive Science Minor* *San Diego, CA*

Relevant Coursework: Data Structures, Systems Programming, Database Management Systems, Web Development

Experience

MC Computer Vision Lab at UCSD | *Research Assistant* **January 2026 – Present**

- Engineered an OpenRooms++ automation pipeline on NRP Kubernetes cluster, integrating Trellis2 and Flux 2.0 to transform coarse meshes into high-fidelity 3D assets.
- Implemented inverse rendering algorithms using SAM3D and Laval Indoor HDR to decompose generated imagery into PBR texture maps (Albedo, Roughness) for photorealistic reconstruction.

Neurocrine Biosciences | *Cloud Engineer Intern* **June 2025 – September 2025**

- Saved $200k in annual OpEx by architecting, building, and deploying an end-to-end serverless containerized data ingestion pipeline on AWS ECS and Lambda, eliminating dependency on third-party vendors.
- Saved 50+ hours per month by developing an AI-driven data pipeline with AWS Bedrock, integrating LLM endpoints with AWS Redshift to convert raw patient data into study-specific narratives, streamlining reporting for study teams.
- Implemented DevOps workflows using Terraform (IaC), AWS Lambda, and multi-environment CI/CD pipelines, collaborating with Jira and Confluence to support Clinical Solution's mission to enhance monitoring of clinical trials.

San Diego County Taxpayers Association | *Software Intern* **October 2024 – January 2025**

- Developed a distributed data ingestion pipeline using Playwright, implementing concurrency controls and proxy rotation to harvest data from dynamic web applications.
- Engineered a high-performance dashboard using React-based libraries, implementing client-side caching and callback optimization to render large-scale datasets with sub-second latency.

Projects

DS3 Internal Membership Portal | *React, TypeScript, Supabase, Postgres, Agile* **July 2025 – Present**

- Directed a 10-person engineering team to architect a React/TypeScript portal for 1,000+ members, replacing manual spreadsheets with a clean, UX-focused web app.
- Engineered CI/CD pipelines to manage a 7,000+ line codebase, automating deployments for rapid bug hotfixes and enforcing strict weekly PR reviews.
- Secured sensitive organization data by building robust Supabase authentication and complex Postgres RLS policies.

Neural Nector | *React, TypeScript, PyTorch, Python, FastAPI, Postgres* **December 2025**

- Developed a full-stack web application serving 100+ users, featuring a React frontend and a RESTful FastAPI backend deployed on cloud infrastructure to challenge users to distinguish between real and GAN-generated flower images.
- Optimized API performance for real-time model inference, managing asynchronous tasks and database transactions to track user game states and leaderboards.

OnlyDance | *React Three Fiber, TypeScript, Python, OpenCV, MediaPipe, FastAPI* **April 2025**

- Won 1st place at DataHacks 2025 (50+ teams) by building an interactive 3D web app that provides real-time motion feedback using browser-based computer vision with MediaPipe.
- Engineered a low-latency feedback loop, processing video frames in the browser and rendering 3D avatars at 60 FPS using React Three Fiber.

YipYap, the Chat App for Yappers | *React, TypeScript, MongoDB, WebSockets* **August 2024**

- Developed a real-time messaging application using Socket.IO for bi-directional event-based communication and Express.js for RESTful API endpoints. Implemented JWT authentication and designed a MongoDB chat and user data.

Technical Skills

Languages: JavaScript, HTML/CSS, Python, SQL, C++, Java, Bash
Certifications: AWS Cloud Practitioner, AWS AI Practitioner, AWS Solutions Architect Associate
Tools: AWS, PostgreSQL, MongoDB, Docker, Terraform, Jira, Confluence, React, Node.js, Express.js, FastAPI, Selenium
Hobbies: 3D Printing, Robotics, Guitar, Rubik's Cube (PB: 26.50s), Pool (Tracked every shot for 700+ games)